Komalpreet Kaur

AI/ML Engineer

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Objective

AI/ML Engineer specializing in multimodal AI, speech intelligence, and cognitive memory systems. Seeking early-stage startup opportunities to build production-grade AI products and intelligent systems.

Projects

Soma (soma.komalpreet.me) *FastAPI, React, LangChain, Neo4j, ChromaDB, SQLite, Docker*

Cognitive memory architecture

- Designed a brain-inspired AI system with multi-layered memory architecture for context-aware reasoning.
- Implemented hybrid RAG pipeline combining ChromaDB and Neo4j with LangGraph for multi-step AI decision-making.
- Built full-stack system with FastAPI and React for consolidation and knowledge refinement.
- Repository Link**: https://github.com/Komalpreet2809/SOMA**

Vanta (vanta.komalpreet.me) *PyTorch, SpeechBrain, FastAPI, Next.js, Docker, Hugging Face*

Speaker-conditioned voice extraction

- Built a Target Speaker Extraction system isolating a target voice from noisy recordings using a 5-second reference clip.
- Trained Conv-TasNet from scratch with SI-SDR loss; paired with frozen ECAPA-TDNN encoder for voice conditioning.
- Deployed FastAPI backend on Hugging Face Spaces and Next.js frontend on Vercel.
- Repository Link**: https://github.com/Komalpreet2809/Vanta**

Specula (specula.komalpreet.me) *Python, FastAPI, PyTorch, HuggingFace, OpenCV*

Deepfake forensic analysis

- Developed a full-stack AI tool using a 5-method forensic pipeline for deepfake image detection.
- Implemented CNN classification with Grad-CAM explainability and analyzers like ELA, FFT, and metadata forensics.
- Designed a weighted scoring system to generate confidence-based authenticity verdicts and enabled batch + PDF reporting APIs.
- Repository Link: **https://github.com/Komalpreet2809/Specula**

Qlothi *Python, FastAPI, PyTorch, HuggingFace, OpenCV, JavaScript*

AI-powered fashion discovery

- Built a Chrome extension for Pinterest enabling AI-based outfit segmentation and product discovery.
- Integrated SegFormer model for clothing detection and automated visual search using Playwright + Google Lens.
- Developed FastAPI backend and interactive UI with filtering features for style, budget, and category-based recommendations.
- Repository Link: **https://github.com/Komalpreet2809/Qlothi**

Skills

Languages:	Python, Java, C++, Bash
Machine Learning & AI:	Supervised & Unsupervised Learning, Deep Learning, Computer Vision, NLP, Feature Engineering
Generative AI & RAG:	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases
Frameworks & Libraries:	PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV, NumPy, Pandas
Backend & APIs:	FastAPI, REST APIs, Model Deployment
Frontend & Tools:	React, JavaScript, HTML, CSS, Chrome Extensions
Databases:	ChromaDB (Vector DB), Neo4j (Graph DB), SQL (SQLite), NoSQL
MLOps & Systems:	Docker, Git, GitHub Actions, CI/CD, Linux
Cloud & Platforms:	AWS, Azure

Education

Lovely Professional University **2023 – 2027**

Bachelor of Technology - Computer Science and Engineering — CGPA: 8.56 *Phagwara, Punjab*

Shri Rajendra Giri Memorial Academy **2021 – 2022**

12th — Percentage: 93% *Kheri, Uttar Pradesh*

Shri Guru Nanak Dev Public School **2019 – 2020**

10th — Percentage: 96.2% *Kheri, Uttar Pradesh*

Note: Links to projects are embedded in their titles. View full portfolio at komalpreet.me